Tommaso de Martino

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WORK EXPERIENCE

European Central Bank – Statistics and Data Science Trainee *Feb 2026 – Present*

- Built and led the development of Python-based pipelines automating the computation of market and credit risk metrics for 100+ European financial institutions, streamlining a process previously run manually
- Sourced, cleaned and validated multi-asset market data (equities, rates, credit) via Bloomberg, producing risk and analytical frameworks used directly by senior risk officers for reporting purposes
- Translated requirements from banking supervisors into quantitative models and automated reporting solutions, working within tight production deadlines on live market data

EDUCATION

MS Economics and Finance 110/110 cum laude – Sapienza University of Rome *Sep 2023 – Oct 2025*

- Focus on Quantitative Methods, Financial Econometrics, Machine Learning in Finance, Mathematics
- Merit scholarship: tuition fee waiver awarded to top 5% of cohort based on GPA
- Thesis: Asset Pricing with Machine Learning

BS Economics – Sapienza University of Rome *Sep 2020 – Oct 2023*

- Focus on Economic Theory, Monetary Policy, Financial Mathematics, Statistics
- Thesis: Dedollarization of the International Monetary System

PROJECTS

Asset Pricing with Machine Learning

- Applied machine learning methods to a large-scale equity dataset of 3M+ firm-month observations covering 23,729 firms and 807 predictors to forecast cross-sectional stock returns
- Neural networks achieved the strongest out-of-sample predictive performance across the model set
- Built long-short equity portfolios from model predictions; best strategy delivered 38.11% annualized return with a Sharpe ratio of 1.98

U.S. Treasury Yield Curve Modeling and Forecasting

- Modeled the U.S. Treasury yield curve using the Diebold-Li three-factor (Nelson-Siegel) model across 8 maturities (3M–10Y), achieving 6bps in-sample RMSE on 437 monthly observations
- Forecasted yield curve factors (level, slope, curvature) via AR(1–5) and VAR(1–5) models across 1/3/6/12-month horizons, using expanding-window out-of-sample validation against a random walk benchmark
- Applied Diebold-Mariano tests to statistically assess forecast accuracy, finding the random walk difficult to beat at short horizons but significant AR/VAR outperformance at longer horizons for several maturities

Volatility Forecasting – Euro Stoxx 50

- Benchmarked GARCH against machine learning models including MLP and Random Forest to forecast Euro Stoxx 50 volatility on 3,763 daily observations
- Engineered 72 cross-asset features across FX, rates, equity indices and commodities to improve volatility forecasting performance
- The best-performing MLP reduced out-of-sample MSE by 20.4% and RMSE by 10.8% relative to the GARCH benchmark, outperforming all other ML specifications tested

Macro Forecasting with Factor Models

- Extracted 7 latent macroeconomic factors from 126 U.S. economic time series (1959–2025) to model broad macroeconomic dynamics
- Produced rolling real-time forecasts for key macro indicators including industrial production and CPI inflation
- The factor model achieved out-of-sample RMSE of 1.37 for industrial production and 0.60 for CPI

SKILLS & INTERESTS

Technical

- Python (advanced): Pandas, Polars, NumPy, scikit-learn, TensorFlow, PyTorch
- SQL; R (advanced); Excel (advanced), Power BI (advanced) Tableau (advanced)
- Experience working with large-scale datasets and server-based computing environments
- Git / GitHub for version control and project documentation

Languages Italian (native); English (fluent); Spanish (fluent); French (intermediate)

Interests Financial markets, Muay Thai and nature